We conducted a parametric analysis on the Lu'an and Ottawa datasets. The effects of key parameters on detection performance are now illustrated in Figures 1 and 2, where each cell in the heatmaps represents the κ coefficient under a specific parameter combination. The results show that κ values vary only slightly within each row, indicating that the proposed CHG method is robust to the number of adjacent neighbors. Meanwhile, as the adjacency neighbors K increases, κ values gradually rise, suggesting that stronger connectivity between neighborhoods improves change detection accuracy.

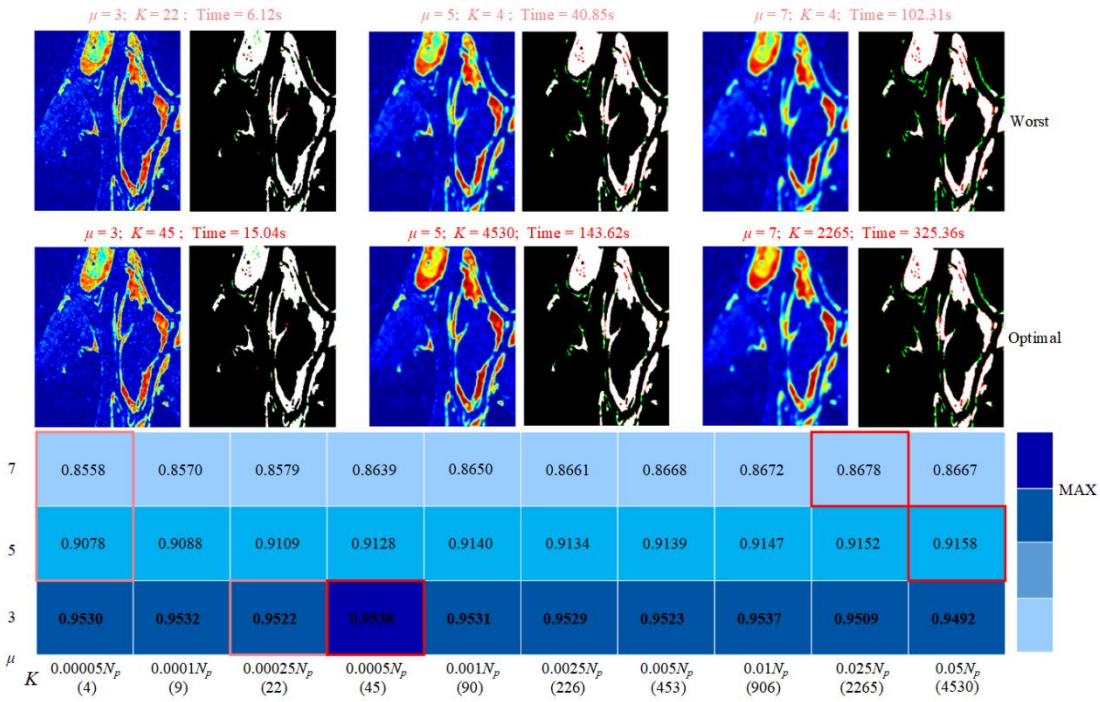

Figure 1. Parameter analysis of the Ottawa dataset

In contrast, κ varies more markedly across columns, revealing that CHG is more sensitive to the patch size μ. Mechanistically, the core of CHG lies in establishing a cross-scale interaction between patch-based attributes and superpixel attributes. The patch size μ directly determines the feature resolution of the PSG. A smaller μ captures finer local texture and edge information, which, when coupled with superpixel attributes, more effectively corrects boundary deviations at the macroscopic scale and preserves small ground objects. Conversely, an excessively large μ leads to oversmoothing of the microscopic representation and weakens the model's ability to resolve complex scenes. The experimental results confirm that a smaller patch size combined with a higher adjacency number helps suppress noise and enhance detail during cross-scale attribute propagation, which is consistent with the parameter selection logic in our experiments.

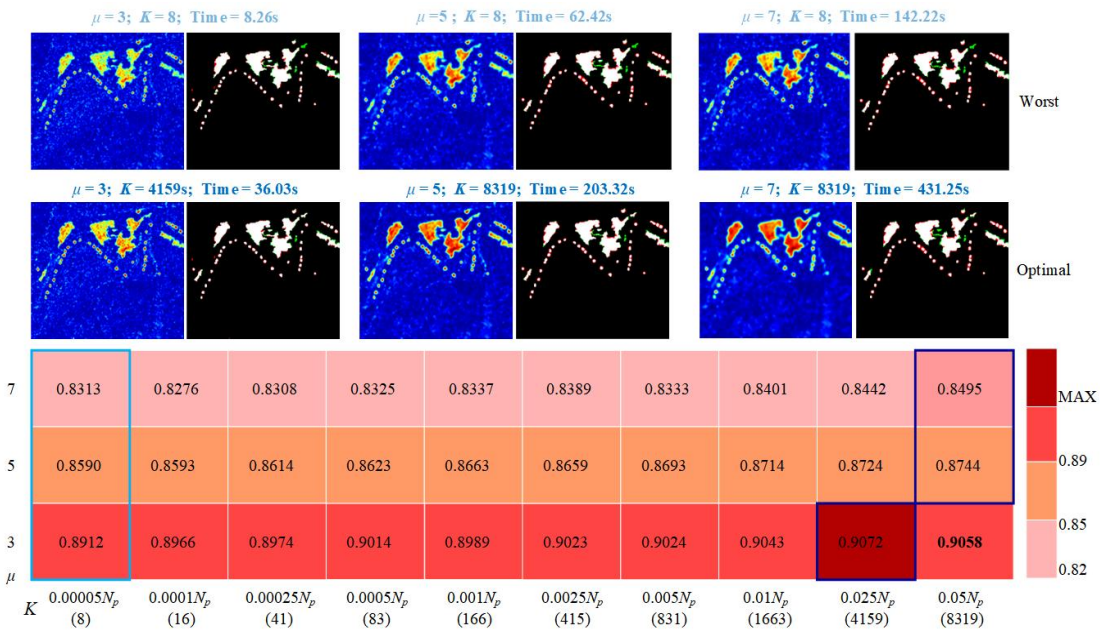

Figure 2. Parameter analysis of the Lu'an I dataset

Furthermore, the top part of the heatmaps visually presents detection results and corresponding computational costs under different parameter settings. Runtime is positively correlated with both the parameters of μ and K. From an algorithmic perspective, the computational burden mainly arises from global graph construction and cross-scale attribute matching. The time complexity of graph construction is $O(N_p^2(\log N_p+\mu^2))$, while that of heterogeneous attribute propagation and matching is $O(N_p^3)$. Based on the analysis, considering the trade-off between accuracy and efficiency, we recommend setting μ in the range [3, 7] and K in [$0.0005N_p$, $0.025N_p$]. This configuration achieves excellent detection performance while keeping computational cost reasonable. Moreover, the combination $\mu = 3$ and $K = 0.001N_p$ is consistent with mainstream settings in related studies, offering good generalizability and robustness. These advantageous parameter setting ranges have been reported in our new manuscript.